

8. Region Merging

Regions: [45, 50] and [52, 55]

(a) Interpretation of merging criteria

Two adjacent regions are merged if their intensity characteristics (e.g. mean values, or the gap between their boundary values) are similar enough — typically, if the difference is within a chosen tolerance, the regions are considered part of the same object and combined.

(b) Justification for the merging approach

Region merging reduces over-segmentation produced by splitting (or by noise), by combining neighbouring regions that are statistically similar into a single, more meaningful region, giving a cleaner final segmentation.

(c) Apply threshold = 5

Boundary values of the two regions are 50 (end of region 1) and 52 (start of region 2). Difference = $|52 - 50| = 2$, which is $\leq$ the threshold of 5, so the merging condition is satisfied.

(d) Computed merged region

Merged region = [45, 50, 52, 55], with combined mean = $(45 + 50 + 52 + 55) / 4 = 50.5$.

(e) Interpretation of result

Since the boundary difference (2) is within the threshold (5), the two regions are close enough in intensity to represent the same underlying object, so they are correctly merged into one region.